Resolution Geometry: Spacetime as the Quotient of a Mesh of Local Systems

Curvature Is the Failure of Local Systems to Form One Flat System; Mesh Closure Below the Obstruction Tower

Matěj Rada
MatejRada@email.cz
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Abstract

We build gravitational geometry with no background spacetime. The only input is a mesh of *local systems*: each is locally flat, and neighbours exchange three kinds of protocol data.

- ruler lengths of shared edges;
- clock ratios;
- frame transports.

Curvature then appears as the failure of the mesh to be one flat system: the holonomy around each hinge, which for surfaces is the angle deficit. We prove and execute five results.

(1) A simply connected mesh is a single flat system iff every hinge holonomy vanishes.

(2) Total holonomy is topological: $\sum \varepsilon_v = 2\pi\chi$ exactly, verified to 10^{-15} on four icospheres.

(3) From ruler data alone, deficit densities converge to the true curvature. On a sphere the maximum error goes $4.8\% \to 0.085\%$. On a Schwarzschild spatial slice, $K = -m/r^3$ is recovered, with median error $1.1\% \to 0.27\% \to 0.066\%$ (second order).

(4) Clock-ratio links along a chain recover the tidal field $-2m/r^3$ to 10^{-6}. This is the electric part of the Riemann tensor, and the lapse constant drops out.

(5) Wall crossing by mesh quotient: link holonomies reproduce the obstruction tower's π_0 and π_1 classes whenever every link step is covered, i.e. below the injectivity bound. Undersampled meshes alias, for example winding $5 \mapsto 1$.

So the *mesh closure* is a cheap, sound closure under coverage, and the obstruction tower becomes its *completionist closure*: it handles uncovered meshes and higher homotopy.

The dynamics is not derived. The discrete Einstein–Hilbert (Regge) action is a selection, consistent with the no-go that the source remainder cannot determine gravity. The upper structure, spacetime, is the quotient of local systems. Its curvature is a gluing defect. Its field equation remains a declared principle.

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1 The mesh contract

Definition 1.1 (Mesh of local systems)**.** A mesh is a cell complex whose top cells are local systems. Each local system is a flat region with its own rulers, clocks and frames. A link between two adjacent local systems carries protocol data:

- the ruler length of each shared edge;
- the clock ratio N_j/N_i, a redshift comparison;
- the frame transport $U_{ij} \in O(n)$ (or a Lorentz element), i.e. how one system's frame is seen by its neighbour.

No ambient spacetime and no coordinates are part of the input.

In the batteries a smooth metric *generates* the synthetic protocol data. The reconstruction never sees that metric; it uses only link data.

2 Curvature as a gluing defect

Theorem 2.1 (Flat iff holonomy-free)**.** *Let a mesh be connected and simply connected. It develops isometrically into a single flat system iff the holonomy of the frame transport around every hinge is trivial.*

Proof. Develop the cells one at a time along a spanning tree of the dual graph. Each cell is flat, and shared edges are identified by their ruler lengths and transports.

The development is consistent iff the composite transport around every dual cycle is trivial. In a simply connected mesh the dual cycles are generated by small loops around hinges. Hence trivial hinge holonomy is equivalent to a global developing map. This is the discrete Frobenius / Cartan argument. □

Theorem 2.2 (Hinge holonomy equals deficit; combinatorial Gauss–Bonnet)**.** *On a surface mesh of flat triangles, the frame carried through the ring of triangles around a vertex v returns rotated by the deficit*

$$\varepsilon_v = 2\pi - \sum \theta.$$

For a closed mesh, $\sum_v \varepsilon_v = 2\pi\chi$ exactly, for any ruler lengths that admit the triangles.

Witness (M1, M2). The deficit totals equal 4π to 10^{-15} on icospheres with 42, 162, 642 and 2562 vertices. The holonomy around one vertex equals its deficit to 10^{-12}; a single flat system gives 0.

Imported Theorem 2.3 (Convergence of piecewise-flat curvature [2])**.** *For a fine and fat (well-shaped) mesh approximating a smooth Riemannian manifold, the deficit measures converge to the scalar-curvature measure. In dimension 2 this is $K\,dA$.*

Witness (M1, M3).

- On the sphere, using ruler (geodesic) lengths only, the maximum relative error of ε_v/A_v against $1/R^2$ is 0.048, 0.012, 0.0033 and 0.00085.

- On the equatorial spatial slice of Schwarzschild with $m = 1$ and $r \in [4, 12]$, rulers integrate the slice metric along the edges. The median relative error against $K = -m/r^3$ is 1.07%, 0.27% and 0.066% for grids of 9^2, 17^2 and 33^2, which is second-order convergence. At $r = 8$ the mesh gives -0.0019544 against an exact -0.0019531.

3 Clocks: the tidal field from clock-ratio links

Proposition 3.1 (Tidal field from clock ratios; M4)**.** *In a static mesh, suppose the links along a radial chain carry ruler lengths ds_k and clock ratios N_{k+1}/N_k. Then the second proper-distance difference of N, divided by N, converges to the radial tidal component $\nabla_s \nabla_s N/N$, which is $-2m/r^3$ for Schwarzschild. The lapse constant cancels.*

At $r = 5$, 7 and 10 the mesh gives -0.0160000, -0.0058309 and -0.0020000 against exact -0.016, -0.0058309 and -0.002.

Rulers give spatial curvature, i.e. structure. Clocks give the tidal field, i.e. the energy footprint. Together they recover the geometric content measured by local protocols, in the duality paper's split. No spacetime is assumed: *the spacetime is the quotient (descent object) of the mesh*, and its curvature is the mesh's gluing defect.

4 Wall crossing by mesh quotient; the tower as completion

Proposition 4.1 (Mesh closure; M6)**.** *On a covered mesh, consider the transporter along a loop. Each link step of a $U(1)$ or $SO(2)$ transport is taken as the shortest angle, which is correct iff the true step is below π. The resulting mesh holonomy equals the obstruction-tower class (the degree) exactly. Without link transport, a pointwise quotient sees nothing. Undersampled meshes alias.*

Witness. Twelve cases with $k \in \{0, 1, 2, 5\}$ and $N \in \{4, 8, 64\}$: every covered case is exact. The uncovered cases alias: $k = 2$ on 4 points reads -2, $k = 5$ on 4 points reads 1, and $k = 5$ on 8 points reads -3.

Corollary 4.2 (Closure hierarchy)**.**

- ***Mesh closure:*** *cost linear in the number of links and plaquettes. It is sound and complete for π_0 and π_1 classes under declared coverage, i.e. a Lipschitz bound times the mesh step stays below the injectivity radius.*
- ***Tower closure*** *(L131): complete for every cell class and every mesh.*

The tower is the completionist closure. It is needed exactly when coverage fails or when classes in π_{r-1}, $r \geq 3$, occur, i.e. on 4-cells and above.

5 What is not derived

No-Go 5.1 (Dynamics is selected, not derived). The mesh gives the *kinematics* of gravity: curvature, tidal field and holonomy. It does not fix how the mesh evolves. The Regge action $\sum_h |h|\, \varepsilon_h$ is the discrete Einstein–Hilbert action [1], and its stationarity gives the discrete field equations. Choosing it is a selection, like Lovelock or the compatibility-plus-H-min selection of linear Einstein in the Both-Sides paper. This is consistent with the print's no-go that the source remainder cannot determine gravity.

Claim	Status	Witness
Curvature is the failure of local systems to form one flat system	LOCKED	Thm 2.1, M2
Total holonomy is topological	LOCKED (exact)	M1
Spatial curvature from rulers alone	LOCKED (convergent)	M1, M3
Tidal field from clock ratios alone	LOCKED	M4
Wall crossing by mesh quotient under coverage	LOCKED	M6
Tower as the completionist closure	LOCKED (hierarchy)	L131, M6
Lorentzian 3+1 Regge mesh, boost holonomies	NOT EXECUTED	—
Field equation of the mesh	SELECTED (Regge), not derived	—

6 Battery and limits

The batteries are `rg_mesh_spacetime.py` (M1–M5, including the canaries: the flat mesh has deficit $< 10^{-9}$, and a cone concentrates $2\pi\delta$ at its apex exactly) and `mesh_vs_tower.py` (M6).

Limits.

- Synthetic link data are generated from smooth metrics.
- Curvature is tested on 2-dimensional (spatial) meshes.
- The tidal field is tested along a static radial chain.
- Lorentzian meshes and dynamics are not executed.
- The convergence theorem is imported [2].

References

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